

Mahindra University, Hyderabad
École Centrale School of Engineering
Minor-I Examinations

Program: B. Tech. Branch: CM Year: II Semester: II
Subject: Number Theory & Cryptography (MA 2209)

Date: 23/02/2026
Time Duration: 90 minutes

Start Time: 02.00 PM
Max. Marks: 17

Instructions:

1. There are 5 questions, all of which are compulsory.
2. Justify your answer wherever required.
3. You can earn up to 2 additional marks beyond the full score by solving Question 3(b).

1. State the following:

[2]

- (a) Kerckhoff's Principle.
- (b) Primitive Root Theorem.

OR

Suppose that $k = (7, 10)$ is a key in an Affine Cipher over $\mathbb{Z}_{36}$. Decrypt the ciphertext 27.

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2. (a) Using the Extended Euclidean algorithm find $u, v \in \mathbb{Z}$ such that $385u + 294v = 14$. [2]
(b) Using the Fast Powering Algorithm compute $3^{248} \pmod{700}$. [3]

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3. (a) Let p be a prime and x be an integer such that $x^2 \equiv 1 \pmod{p}$. Show that either $x \equiv 1 \pmod{p}$ or $x \equiv -1 \pmod{p}$. [1]
(b) Let p be a prime such that $q = \frac{1}{2}(p-1)$ is also a prime. Suppose that g is an integer satisfying $g \not\equiv \pm 1 \pmod{p}$ and $g^q \not\equiv 1 \pmod{p}$. Prove that g is a primitive root modulo p . [1] + [2 Bonus Marks]

4. Bob's RSA public key cryptosystem has modulus $n = 221$ and encryption exponent $b = 11$. Alice sends Bob the ciphertext $c = 100$. Unfortunately, Bob has chosen too small a modulus. Help Eve in decrypting Alice's message. [4]
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5. Solve the following simultaneous systems of congruences using Chinese Remainder Theorem.

$$x \equiv 3 \pmod{9}, \quad x \equiv 2 \pmod{7}, \quad \text{and} \quad x \equiv 5 \pmod{8}.$$

[4]